



A Wearable Ergonomic Risk Awareness System for Occupational Settings

Group 6 - Pervasive Computing – April 2026

Intro & Motivation

- Over **500,000 workers** suffer work related back and spine injuries every year (HSE, 2024)
- **60-80% of wearable devices are abandoned within 2-4 weeks** workers ignore devices that vibrate too much or too early
- Current systems fail not because of bad sensors but because of **bad feedback approach (timing, vibration duration, etc)** causing alarm fatigue.
- **Our goal:** detect bad posture in real-time and deliver feedback only after 5 seconds of sustained bad posture, reducing alarm fatigue while still protecting the worker.

System Architecture

Processing Unit

- ESP32-S3 microcontroller

Communication

- I2C protocol
- TCA9548A I2C multiplexer
- Sequential sensor selection

Sensors

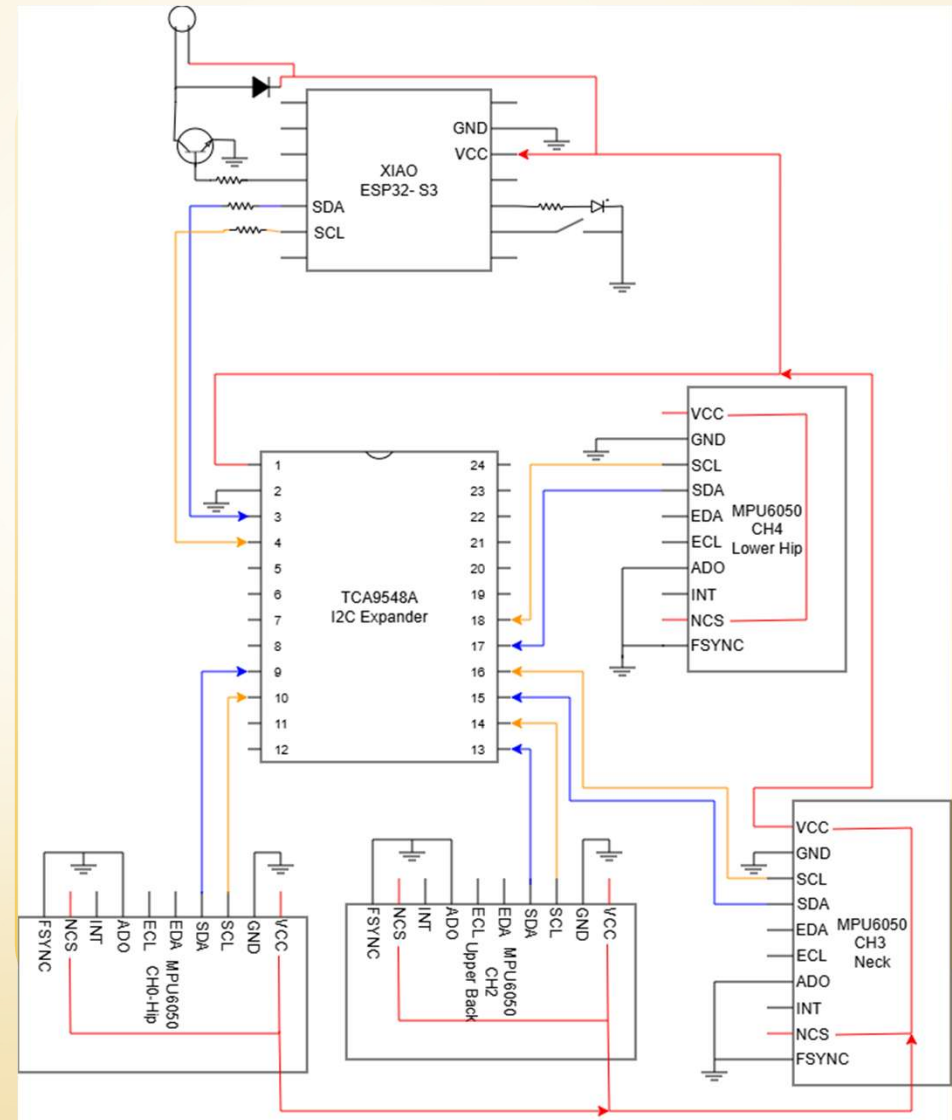
- 4 × MPU6050 IMU 6-axis
- placement: hip, LB, UB, neck

Central Logic

- FSM
- ML – Feed Forward NN Classifier

Circuit Design

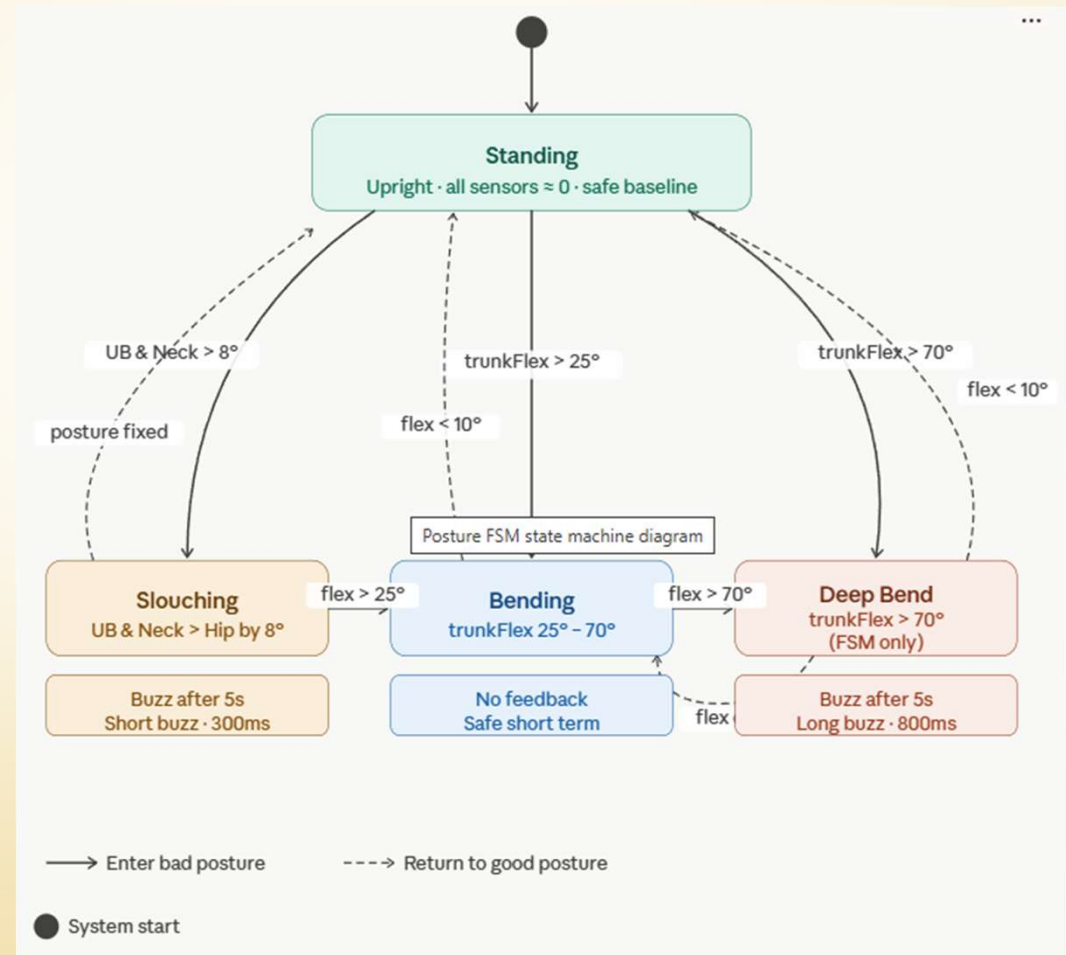
- **Input:** Sensors -> I2C multiplexer -> ESP32-S3
- **Output:** ESP32 -> vibration motor and LED (+ transistor for safety)
- **Reliability and safety:** flyback + diode + resistor

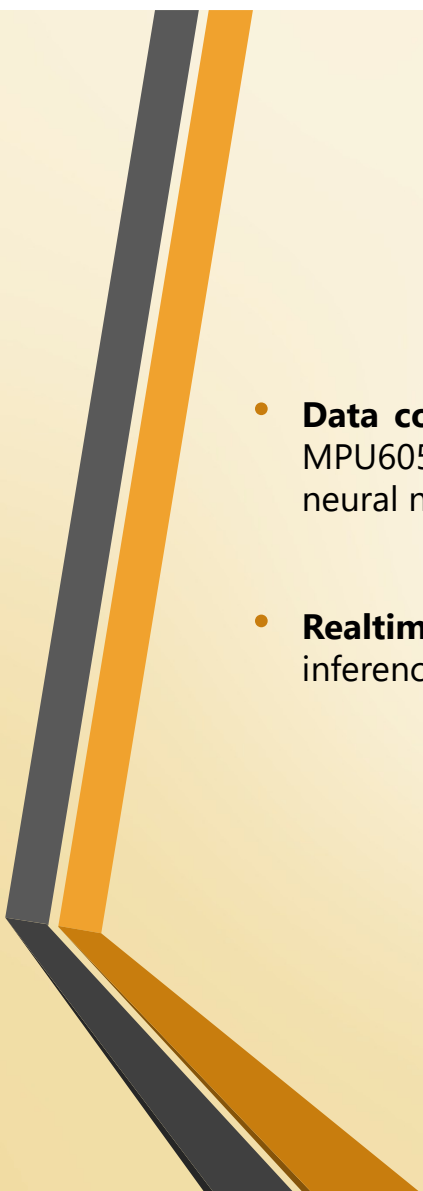


Central Logic: FSM

- **Standing** - upright, all sensors near zero. Safe baseline.
- **Slouching** - upper back rounds and head pushes forward while standing. Hip stays upright.
- **Bending** - leaning forward from the lower back. Safe short term, no feedback.
- **Deep Bend (only FSM)** - extreme forward lean. Dangerous for spinal discs.
- **Feedback Rule:** Bad posture (Slouching or Deep Bend) held for 5 seconds triggers motor buzz. Deep Bend = long buzz (800ms). Slouching = short buzz (300ms).

More context in the Live Demo.

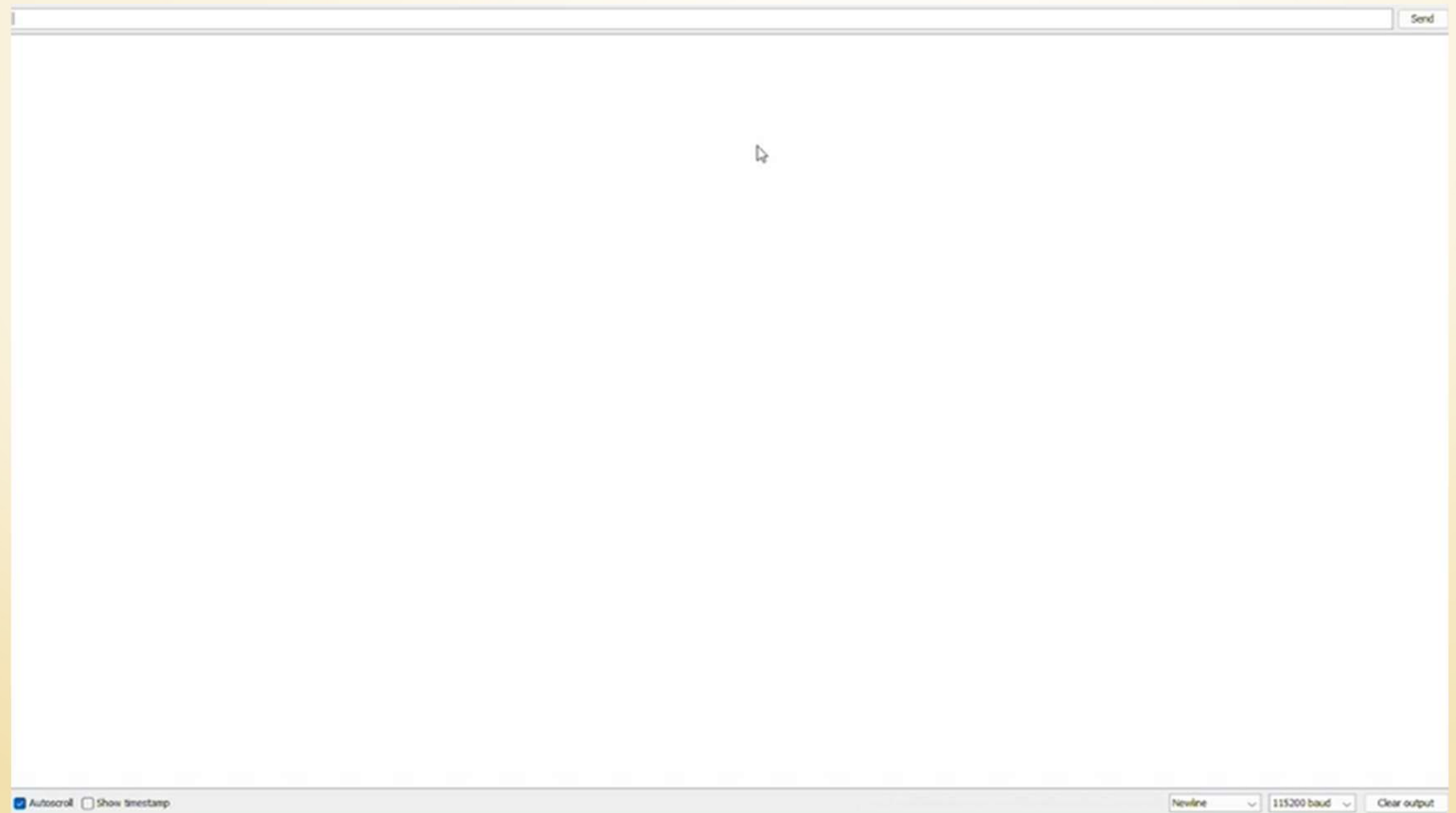




Machine Learning Approach

- **Data collection and training:** The workflow in Edge Impulse includes collecting motion data from the MPU6050 sensors, labeling it into different posture categories, and training a Feed-forward, classifier, neural network model.
- **Realtime deployment:** The trained model is then optimized and deployed to the ESP32-S3 for on-device inference.

Results





Live Demo

For the love of god, just work :)



Thanks!